

RYBINSK, Russian Federation

WMO: 272250

Lat: 58.100N

Lon: 38.683E

Elev: 345

StdP: 14.51

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-10.2	-20.3	1.8	-15.9	-14.3	2.6	-9.9	20.1	20.5	18.0	22.3	3.4	310	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	83.4	69.4	80.0	67.7	76.7	65.9	71.0	80.4	69.1	77.8	67.2	74.9	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	102.9	76.5	65.7	96.1	74.0	63.9	90.1	72.3	35.0	80.5	33.4	78.2	31.8	75.2	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	16.1	14.3	-22.2	87.8	5.7	3.4	-26.3	90.2	-29.7	92.2	-32.9	94.1	-37.0	96.5
			-22.4	73.1	5.6	3.0	-26.5	75.3	-29.8	77.1	-33.0	78.8	-37.1	81.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	16.3	16.8	26.4	40.9	53.2	61.6	66.6	61.8	52.0	40.9	29.2	21.2	(6)
(7)		DBStd	19.97	13.19	13.78	9.45	8.91	8.68	6.80	5.68	6.13	6.59	7.56	9.74	12.62	(7)
(8)		HDD50	4941	1043	929	733	300	65	2	0	1	53	296	626	893	(8)
(9)		HDD65	9061	1508	1349	1198	724	377	147	48	136	393	748	1075	1358	(9)
(10)		CDD50	1545	0	0	0	25	165	348	516	367	112	12	0	0	(10)
(11)		CDD65	195	0	0	0	0	12	44	99	38	2	0	0	0	(11)
(12)		CDH74	1335	0	0	0	4	110	331	635	247	8	0	0	0	(12)
(13)		CDH80	294	0	0	0	0	26	72	139	57	0	0	0	0	(13)
(14)	Wind	WSAvg	6.3	6.2	6.4	6.3	6.1	6.7	5.9	5.3	5.6	6.3	7.1	6.7	6.7	(14)
(15)	Precipitation	PrecAvg	26.0	1.9	1.5	1.4	1.2	2.1	2.9	2.9	3.2	2.6	2.3	2.1	2.0	(15)
(16)		PrecMax	33.7	2.7	3.0	2.4	2.8	4.9	6.7	5.9	7.0	5.0	6.0	4.3	3.1	(16)
(17)		PrecMin	19.6	0.7	0.5	0.3	0.5	0.6	0.2	0.3	0.7	0.9	0.8	0.2	0.6	(17)
(18)		PrecStd	3.9	0.6	0.7	0.5	0.6	0.9	1.5	1.6	1.7	1.1	1.1	1.1	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	39.4	50.4	72.1	83.5	86.2	88.1	86.0	74.3	64.0	49.4	44.8	(19)
(20)			MCWB	36.4	36.2	42.2	56.4	65.9	69.0	72.3	71.5	63.3	55.0	46.8	42.6	(20)
(21)		2%	DB	35.7	36.8	44.7	65.8	76.4	81.8	83.7	79.9	69.0	57.1	45.6	40.4	(21)
(22)			MCWB	34.2	34.2	38.6	53.6	62.7	68.5	70.0	67.5	61.0	51.8	43.5	38.7	(22)
(23)		5%	DB	34.3	35.1	40.7	60.1	71.4	77.9	80.6	76.0	65.4	53.6	42.9	37.2	(23)
(24)			MCWB	33.0	33.0	36.1	49.9	59.5	66.6	68.5	65.6	58.5	49.8	41.1	35.5	(24)
(25)		10%	DB	32.9	33.7	38.0	55.2	67.2	73.8	77.4	72.5	62.0	51.1	40.5	34.9	(25)
(26)			MCWB	31.7	32.0	34.6	47.0	56.8	63.8	66.8	63.8	56.5	48.1	38.8	33.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	36.9	43.2	58.4	69.3	73.0	74.3	73.3	66.0	56.2	47.5	43.4	(27)
(28)			MCD B	37.5	38.5	49.5	71.2	80.2	82.8	85.3	83.2	72.3	62.1	49.2	44.9	(28)
(29)		2%	WB	34.5	34.8	39.4	53.8	63.8	70.0	71.3	69.3	62.1	52.8	43.9	38.8	(29)
(30)			MCD B	35.3	36.3	43.1	64.4	74.8	79.8	81.1	76.9	67.3	56.3	45.2	40.3	(30)
(31)		5%	WB	33.2	33.4	36.9	50.7	60.8	67.6	69.6	66.9	59.4	50.4	41.5	35.7	(31)
(32)			MCD B	34.1	34.8	39.7	58.6	69.1	75.8	78.7	74.3	64.1	53.1	42.7	36.9	(32)
(33)		10%	WB	31.6	32.1	34.9	47.5	58.1	65.1	67.8	64.8	57.0	48.4	38.8	33.7	(33)
(34)			MCD B	32.8	33.8	37.5	54.6	65.6	72.2	75.8	71.2	61.7	50.9	40.3	34.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.5	11.8	13.4	15.3	15.4	14.4	14.5	14.1	12.2	8.7	6.7	7.9	(35)
(36)			MCD BR	7.3	9.1	15.1	23.5	22.6	20.3	19.8	20.4	18.1	13.4	8.4	6.4	(36)
(37)			MCWBR	7.2	8.0	10.2	13.7	11.4	9.8	9.1	9.7	10.2	8.6	7.7	6.1	(37)
(38)		5% WB	MCD BR	7.2	8.5	12.3	21.5	20.0	18.5	17.6	18.2	15.7	12.3	7.9	5.9	(38)
(39)			MCWBR	7.2	8.0	8.8	13.3	10.8	9.7	8.8	9.6	9.7	8.6	7.6	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.322	0.342	0.393	0.374	0.367	0.403	0.404	0.367	0.343	0.307	0.268	(40)	
(41)		taud	2.075	2.063	2.110	2.187	2.383	2.422	2.344	2.319	2.398	2.358	2.185	2.056	(41)	
(42)		Ebn at Noon	169	215	249	257	273	276	262	252	244	213	159	133	(42)	
(43)		Edh at Noon	17	28	36	39	34	34	36	34	27	21	16	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	140	395	854	1243	1677	1748	1712	1326	788	348	122	74	(44)	
(45)		RadStd	23	74	107	134	129	136	201	102	101	58	20	17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	-400	N/A	N/A

Nomenclature: See separate page